

vector architectures, instruction set extensions

variety of designs that exploit **data parallelism** using some interpretation of “**SIMD**” to speed-up operations on **vectors** of data

Flynn’s taxonomy classifying computer architectures:

“**SISD**” (“**single instruction stream, single data stream**”) i.e. conventional uniprocessors

“**MIMD**” (“**multiple instruction stream, multiple data stream**”) i.e. multiprocessors

“**SIMD**” (“**single instruction stream, multiple data stream**”)

“**MISD**” (unused)

vector instruction set extensions in x86

- **MMX** (“**MultiMedia eXtension**”), **SSE** (“**Streaming SIMD Extensions**”), **SSE2**, **AVX** (“**Advanced Vector Extensions**”), **AVX2**, **AVX-512**, **AVX10**, ...
- use large registers (e.g., 512 bits in **AVX-512**) that can each store a short vector of operands (e.g., in **AVX-512**, 16 single precision floating point values)
- instructions provided for operations on these vectors (e.g., single instruction can do pair-wise addition of two vectors each with 16 floating point values)
- **AMX** (“**Advanced Matrix Extensions**”) extends idea to *matrices* (have 2-d registers called *tiles*) ... important application domain is deep-learning training & inference

vector instructions in RISC-V (extension `V')

portion of [example program](#) from course Canvas site:

```
...
vec1:    .space 40
vec2:    .space 40
vec3:    .space 40
...
# read 10 single precision floating point values from user,
# store in memory at "vec1"; ditto for "vec2"
...
la s1, vec1
la s2, vec2
la s3, vec3
li s0, 10
vprod:   vsetvli t0, s0, e32      # sets "vtype" and "vl" registers
# "e32" means "element width" is 32 bits
# total number of vector elements remaining to be
#     processed given in s0
# number of vector elements that will be processed in
#     current iteration ⇒ t0 (typically, = min[VLMAX, s0])
...
vle32.v v0, (s1)
vle32.v v1, (s2)
vfmul.vv v2, v1, v0
vse32.v v2, (s3)
slli t1, t0, 2                  # why?
add s1, s1, t1
add s2, s2, t1
add s3, s3, t1
sub s0, s0, t0
bnez s0, vprod
```

GPUs (“graphics processing units”)

originated from merging of specialized hardware units used for graphics

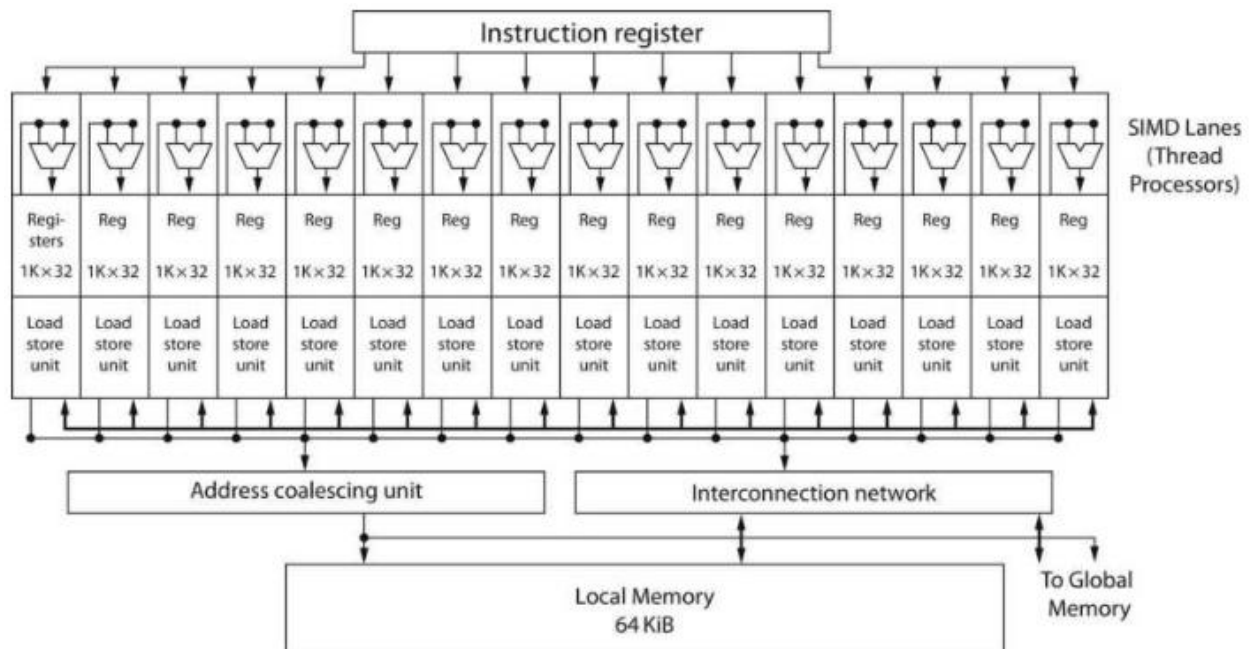
now used for many other computationally-intensive applications (since GenAI, NVIDIA GPUs, e.g., in exceptionally high demand ... 10yr stock return Feb 2016-2026 > 24,000%!)

exploit **data parallelism**

representative high-level design:

- GPU has multiple **multithreaded SIMD processors**
- each SIMD processor has multiple ALUs that execute identical instructions (on different data) in lock-step
- *block scheduler* assigns blocks of threads to the SIMD processors; *thread scheduler* within each SIMD processor can use fine-grained multithreading idea

(simplified picture of a SIMD processor; from text Fig. 6.9)



specialized processing accelerators

targeted at particular applications such as **image and video processing, machine learning, ...**

accelerators for AI applications, sometimes termed “neural processing units” (NPU), have become widely used

- found in smartphones, personal computers, cloud computing systems, ...
- an example is Google's “[tensor processing unit](#)” (TPU) ([7th gen TPU \(“Ironwood”\) announced last year](#))

TPUv1

- specialized for matrix processing (as needed for neural network training/inference ...)
- matrix multiply unit contains array of 256 x 256 (65,536) multiply-accumulators ... SIMD parallelism!
- dedicated memories
- smaller data types (8 and 16 bit integers)
- power efficient ... included architectural units and features tailored for application domain of interest!

also much current interest in specialized CPUs for supporting “agentic AI” (to support planning, orchestration, tool use ..., e.g. NVIDIA Vera CPU, ARM AGI CPU)

what's next?

“multi-chip packages”, denser packing of compute nodes (e.g. NVIDIA Blackwell and new Vera Rubin GPUs, liquid-cooled “rack-scale systems”)

wafer-scale integration (e.g. Cerebras’s “Wafer-Scale Engine”; WSE-3 chip is 21.5 x 21.5 cm and includes 900,000 cores)

ever bigger systems (e.g. xAI’s “Colossus”, currently > 550,000 NVIDIA GPUs, plans for 1 million)

photonic computing? (e.g. Lightmatter’s photonic AI processor)

“software-defined” dataflow? (e.g. NextSilicon’s “Maverick-2”)

new VLSI technologies?

neuromorphic computing?

quantum computing?

... and what new applications will be enabled?

AGI? ASI?